γδ T cells: Major advances in basic and clinical research in tumor immunotherapy

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Abstract

γδ T cells are a kind of innate immune T cell. They have not attracted sufficient attention because they account for only a small proportion of all immune cells, and many basic factors related to these cells remain unclear. However, in recent years, with the rapid development of tumor immunotherapy, γδ T cells have attracted increasing attention because of their ability to exert cytotoxic effects on most tumor cells without major histocompatibility complex (MHC) restriction. An increasing number of basic studies have focused on the development, antigen recognition, activation, and antitumor immune response of γδ T cells. Additionally, γδ T cell-based immunotherapeutic strategies are being developed, and the number of clinical trials investigating such strategies is increasing. This review mainly summarizes the progress of basic research and the clinical application of γδ T cells in tumor immunotherapy to provide a theoretical basis for further the development of γδ T cell-based strategies in the future.

Keywords: Gamma-delta T-cells; T cell receptor; CD1a; Tumor immunotherapy; Receptors, chimeric antigen; Vδ1 cells; αβ T cells

Introduction

γδ T cells are a unique type of innate immune T cell that express a γδ T cell receptor (TCR). Mature γδ T cells are highly abundant in epithelial mucosal tissues, such as the skin and digestive, respiratory, and reproductive tracts, and these are areas with high incidence rates of various solid tumors.[1–3] Lymphocytes that infiltrate tumor tissue in patients also include γδ T cells, which can directly recognize tumor-associated antigens without major histocompatibility complex (MHC) restriction and achieve the rapid, direct killing of tumor cells. In addition, γδ T cells secrete various cytokines that regulate antitumor immune responses via cellular interactions.[4–7] In contrast to the challenges encountered by many adoptive cell therapies, γδ T cells have no autologous limitations, can be derived from healthy donors, and have a simple process of expansion that results in powerful killing functions *in vitro*.[8,9] Therefore, γδ T cells are considered to be among the best candidate cells to use in immunotherapy.[10]

Current therapies that are based on γδ T cells use two main mechanisms. One is the selective amplification of γδ T cells *in vivo* using antibodies or bisphosphonate antigens. The other is adoptive cell therapy, and in this approach, tumors are treated either by γδ T cells that are expanded *in vitro* or by injection with allogeneic γδ T cell products, including natural γδ T cells and some genetically engineered γδ T cells. In recent years, γδ T cells have become the "new dark horse" in the field of immune cell therapy.[10–14] Many countries, such as the United States, the United Kingdom, and Singapore, are actively pursuing the distribution of allogeneic natural γδ T cell products and genetically engineered γδ T cell products. Several γδ T cell-based therapeutic methods have shown positive progress.[15] Relatively speaking, the track of γδ T cell research in China is in its infancy, with a few companies currently in the field, most of

Access this article online

Quick Response Code:

Website: www.cmj.org

DOI: 10.1097/CM9.0000000000002781

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Chinese Medical Journal 2024;137(1)

Received: 03-05-2023; **Online:** 18-08-2023 **Edited by:** Jing Ni

which are in the early clinical stage; however, γδ T cell research is gaining momentum.

Because γδ T cells are rare in the peripheral circulation and lymphoid organs, they have not been studied in-depth, and many fundamental questions remain unanswered. However, with the continuous development of γδ T cells for clinical application, basic research on γδ T cells has shown some achievements. The field of γδ T-cell immunotherapy has both opportunities and challenges that need to be addressed through further research.[16] This review mainly summarizes the latest progress in basic studies and clinical trials of γδ T cells in 2022 and provides a theoretical basis for the further development of γδ T cell-based strategies in the future.

Basic Research on γδ T Cells in Tumor Immunity

γδ T cells are a small group of innate immune cells, so basic research on γδ T cells has not received substantial attention. Many hypotheses about the mechanisms of γδ T cells are simply based on what is known about natural killer (NK) cells or αβ T cells. In addition, most of the relevant research has been conducted in mice, but γδ T cells from mice and humans are quite different, and conclusions from one species cannot be applied mechanically to the other species. The current status of γδ T cell research is that clinical studies are being conducted first, while basic research is lagging.[15] In recent years, due to the unique advantages of γδ T cells in tumor immunotherapy, an increasing number of basic studies have focused on the mechanisms underlying antigen recognition, activation, functional depletion, and tumor immune response. γδ T cells are highly cytotoxic, but there is a high degree of safety. Therefore, γδ T cells have been considered a good alternative therapeutic approach for cancers.[15,17–19] However, like many immune cells, γδ T cells play a dual role in tumor immunity. Some γδ T-cell subpopulations, particularly those that secrete interleukin-17 (IL-17), suppress the immune response and promote tumor immune escape.[20–26]

Mechanism underlying ligand recognition by γδ T cells

Cluster of differentiation (CD) 1a is a ligand of Vδ1⁺ γδ T cells

γδ T cells are amplified in response to recognition of γδ TCR ligands, so γδ T cells not only have the characteristics of innate immune cells but also perform adaptive immune functions.[6,27] However, to date, the identification of γδ TCR ligands is still very limited, and the mechanism by which γδ T cells recognize these ligands is unclear, limiting an in-depth understanding of the role of γδ T cells in the immune response.[28]

CD1a is a monomer antigen-presenting molecule that is expressed by dendritic cells and presents lipid antigens to αβ T cells.[29] In July 2022, Wegrecki *et al*[30] at the Institute of Biomedical Discovery at Monash University in Australia published a study in *Nature Communications*. They used a CD1a tetramer to demonstrate that CD1a was a ligand for $V\delta1^+$ γδ T cells. The results of functional experiments showed that the $V\delta1^+$ γδ TCR binded to CD1a in a lipid antigen-independent manner. The crystal structure further indicated that the $V\delta1^+$ γδ TCR binded to the farthest side of the β-substrate structure from the antigenic binding groove of CD1a in a direction that was parallel to the major axis. In addition, the $V\delta1^+$ γδ TCR recognized the heavy chain and β2 microglobulin of CD1a molecules in a manner that was different from other previously observed mechanisms of γδ TCR recognition.

This study revealed that $V\delta1^+$ γδ TCR recognized CD1a via a "sideways" and lipid antigen-independent manner, which not only revealed an "end-to-end" antigen-recognition mechanism that was different from that of αβ TCRs, which were usually directly exposed to antigens, but also lay the foundation for the clinical use of $V\delta1^+$ T cells to target CD1a-associated tumor lipid antigens.

γδ T cells perform tumor surveillance by recognizing selection and upkeep of intraepithelial T cells 1 (Skint1)

γδ T cells are located in epithelial mucosal tissues, such as the skin and intestinal and respiratory mucosa. They play an immune surveillance role, sensing foreign infections and changes in the body.[31–33] However, the mechanism by which γδ T cells differentiate between healthy and abnormal cells remains unclear.

In March 2022, a study published in *Nature Immunology* by Mckenzie *et al*[34] from the Francis Crick Institute in the UK revealed the mechanism by which γδ T cells played a surveillance role in tumor immunity.

Researchers analyzed interactions between γδ T cells in the skin and other cells that formed the structure of the mouse skin. They found that γδ T cells sensed the Skint1 molecules that were present in healthy skin cells. When skin was exposed to common forms of stress, including ultraviolet radiation and harsh chemicals, γδ T cells sensed changes in Skint1 molecules, identified these environmental challenges, and helped the skin recover. When binding to Skint1 was blocked, γδ T cells were unable to function properly.

In addition to Skint1 in the skin, Skint1-like molecules were expressed in different tissues.[35] These findings might help to explain the molecular mechanisms by which γδ T cells maintained and protected multiple parts of the body.

$V\delta1^+$ T cells recognize human leukocyte antigen (HLA)-DR molecules

In healthy adults, $V\delta1^+$ T cells account for only one-third of circulating γδ T cells, and they are found primarily in peripheral tissues, including the intestinal epithelium, dermis, spleen, and liver. $V\delta1^+$ T cells are the major γδ T-cell subpopulation and contribute to tissue homeostasis.[36] During cytomegalovirus (CMV) infection and to a lesser degree Epstein–Barr virus (EBV) infection, $V\delta1^+$ T cells are mediators of the antiviral immune response.[37–39] In September 2022, Deseke *et al*[40] from Hannover Medical School in Germany published a paper in *Journal of Experimental Medicine*. Their latest study titled "A CMV-induced adaptive human Vδ1+ γδ T cell clone recognizes HLA-DR," has also been published. This group dynamically observed changes in the γδ *TCR* gene pool during CMV infection and found that single Vδ1+ T cell clone prolifically expanded during viral infection. To determine whether this amplification was random or represented a property of the γδ T-cell TCR-dependent adaptive immune response, they further used clustered regularly interspaced short palindromic repeats (CRISPR)/Cas9-mediated screening to collect information about the ligand that was recognized by this TCR. The results showed that the strongly amplified Vγ3Vδ1+ TCR could directly interact with the HLA-DR MHC II complex during CMV infection.

Based on the recognition of MHC I-related molecules by Vδ1+ T cells, Vδ1+ T cells were further proposed to directly recognize MHC II molecules. Since the expression of MHC II molecules was strongly upregulated in the inflammatory environment, this study also revealed the mechanism by which γδ T cells mount an inflammation-induced MHC-dependent immune response.

Mechanism underlying the antitumor effect of γδ T cells

γδ T cells play a key role in responses against tumor cells that lack the human leukocyte antigen type I (HLA I)

DNA mismatch repair defects cause tumor cells to exhibit many characteristics that normal cells do not have.[41] Normally, T cells recognize these abnormal cells based on antigen presentation by HLA I molecules. However, some tumor cells lack HLA I molecules to avoid being killed by T cells.[42,43]

de Vries *et al*[44] from Leiden University Medical Center in the Netherlands published a new study in *Nature* that revealed that γδ T cells played an important role in identifying and removing tumors that were invisible to the immune system's conventional defense mechanisms. They observed that patients with β2-microglobulin deficiency responded to immune checkpoint inhibitors, and this effect was primarily associated with tumor-infiltrating γδ T cells. These γδ T cells consisted mainly of Vδ1 and Vδ3 subpopulations and expressed high levels of programmed cell death-1 (PD-1) and other activation markers, including cytotoxic molecules, and a wide range of killer cell immunoglobulin-like receptors. After PD-1 and (cytotoxic T lymphocyte-associated antigen-4 (CTLA-4) inhibition, the frequency of γδ T-cell infiltration in the tumor was greatly increased. The killing of tumor cells by these γδ T cells was independent of HLA I molecules and thus improved the response of HLA-negative patients to immune checkpoint inhibitors.

Although γδ T cells have been widely used, basic research on these cells is still limited. This study confirmed the great potential of γδ T cells for use in cancer immunotherapy and supported the use of γδ T cells to develop novel therapies for invisible cancers.

Vγ9Vδ2 T cells remain in the liver and target hepatocellular carcinoma

Globally, hepatocellular carcinoma (HCC) is the fourth leading cause of cancer-related death, the sixth leading cause of new cancer cases, and the second leading cause of death after pancreatic cancer. At present, the prospects of HCC immunotherapy are very promising, and tumor immunotherapy has become the standard treatment for advanced HCC. However, there are still many patients who do not respond to immunotherapy, and how to enhance the antitumor effects of memory T cells with strong HCC responsiveness and tissue residency is an urgent problem.[45–47]

In March 2022, Zakeri *et al*[48] from University College London, UK, published research in *Nature Communication*. They found that γδ T cells infiltrated HCC in high numbers and had the potential for HLA-independent tumor reactivity; they also showed that the number of γδ T cells was positively correlated with HCC survival. In addition, peripheral blood Vγ9Vδ2 T cells remained in the liver as they passed through and did not escape from liver vessels. These Vγ9Vδ2 T cells targeted the accumulation of isopentenyl-pyrophosphate in HCC cells, exerting in potent cytotoxic effects and continuously producing antitumor cytokines.

Based on these results, the authors proposed a strategy of Vγ9Vδ2 T cell amplification *in vivo* by combining amino-bisphosphonates or the adoptive transfusion of Vγ9Vδ2 T cells from tumors *in vitro*, which could effectively replenish depleted γδ T cells. Further study of the recognition mechanism and the role of γδ T cells in the immune response will enhance our understanding.

Vδ1+ T cells in the lung are critical for the survival of lung cancer patients

Tumor-infiltrating γδ T cells have been shown to be the immune cells that most are positively associated with patient survival in all cancers.[49,50] When examining brain tumor samples from patients, γδ T-cell infiltration was shown to be the strongest predictor of survival, while αβ T-cell infiltration was negatively associated with survival.[51]

Non-small cell lung cancer (NSCLC) accounts for approximately 85% of all lung cancer cases and has a five-year survival rate of only approximately 20%, so there is an urgent need to develop new therapies.[52,53] In May 2022, hundreds of researchers from more than 50 research institutions jointly published the latest research results in *Nature Cancer*, which revealed the important role of Vδ1+ T cells in lung cancers.[54] The research team analyzed lung tumor tissue samples and non-cancer lung tissue samples from 25 NSCLC patients. They found that Vδ1+ T cells were present in both lung tumors and non-cancer lung tissues and that the presence of Vδ1+ T cells in lung tumors was associated with the maintenance of sustained remission after surgery. Notably, the presence of Vδ1+ T cells in non-cancerous lung tissues was a stronger predictor of sustained remission, suggesting that Vδ1⁺ T cells continue to function even after tumor resection. They also found that Vδ1⁺ T cells had stem-like characteristics that allowed them to self-renew in lung tumors, and this property was important for the long-term control of lung cancer.

Immunotherapy is an important tumor treatment, but lung cancers are often limited and difficult to predict with respect to these treatments. By studying how Vδ1⁺ T cells work in cancer and therapy, we can identify potential new ideas for improving the effectiveness of γδ T-cell immunotherapy and offer hope to patients.

Mechanism underlying the protumor effects of γδ T cells

Tumor cells achieve immune escape by expressing butyrophilin-like 2 (BTNL2) and interacting with IL-17A⁺ γδ T cells

In recent years, immune checkpoint inhibitors have come to the forefront of tumor immunotherapy, including CTLA-4 and PD-1/programmed cell death-ligand 1 (PD-L1) antibodies, and have shown good efficacy against various tumor types.[55] However, the overall response rate of tumor patients to current immune checkpoint inhibitor therapy remains low, suggesting that other important immunosuppressive molecules or mechanisms may be present.[56–58] Therefore, the search for new tumor immunosuppressive molecules and the development of agents that inhibit these new targets are among the most important areas of research in the field of tumor immunity.

In February 2022, Du *et al*[59] from Huazhong University of Science and Technology published research in *Nature Communications*. BTNL2 is a type I transmembrane immune regulatory protein that is highly expressed in the gut. This group found that BTNL2 inhibited antitumor immunity by acting on γδ T cells in the tumor microenvironment to promote IL-17A secretion and recruit myeloid suppressor cells, resulting in decreased cytotoxic CD8⁺ T-cell infiltration. The anti-BTNL2 antibody had a significant therapeutic effect on various mouse tumors and exhibited a certain synergistic therapeutic effect with the anti-PD-1 antibody. Additionally, the expression of BTNL2 in tumors was highly correlated with the prognosis of patients and the infiltration of IL-17A-expressing γδ T cells, and the expression level of BTNL2 in almost all tumors was significantly higher than that in control tissues.

These results suggested that BTNL2 was an attractive target molecule for tumor immunity and that antibody-inhibitor drugs that target BTNL2 had the potential to enhance the efficacy of currently available tumor immunotherapeutic strategies; these findings might provide alternative treatment options for patients who are currently resistant to tumor immunotherapy.

γδ T cells that secrete IL-17A are resistant to immune checkpoint inhibitor therapy

Breast cancer is one of the main malignant tumors that threaten women's health. When breast cancer spreads to other parts of the body, it becomes much more difficult to treat.[60,61] A study published in *Journal of Experimental Medicine* in December 2022 revealed the mechanism by which γδ T cells impaired the efficacy of immunotherapy in breast cancer.[62] γδ T cells that secreted IL-17A in mice were found to mainly include Vγ6⁺ tissue-resident cells and Vγ4⁺ circulating cells. In the lung of tumor-bearing mice, Vγ6⁺ cells constitutively expressed high levels of PD-1, while Vγ4⁺ cells upregulated T cell immunoglobulin domain and mucin domain-3 (TIM-3) expression in response to tumor-derived IL-1β and IL-23. Immune checkpoint inhibitors against PD-1 or TIM-3 increased the number of Vγ6⁺ and Vγ4⁺ cells in the lung, respectively, causing breast cancer to spread to the lung of the treated mice. In the model of breast cancer that was insensitive to checkpoint inhibitors, knocking out γδ T cells elicited response to anti-PD-1 and anti-TIM-3 immunotherapy.

This study revealed the resistance of γδ T cells that secreted IL-17A to immune checkpoint inhibitor therapy. PD-1 and TIM-3 antibodies differentially controled IL-17A-secreting γδ T cells in the lungs of patients with homeostasis and cancer, making breast cancer more likely to spread to the lung tissue; these results suggested that immune checkpoint inhibitor therapy in combination with targeting γδ T cells and IL-17A molecules might be more effective in preventing the lung metastasis of breast cancer.

Regulation of antitumor immunity by γδ T cells

Intestinal antitumor and protumor γδ T cells are associated with different TCR gene usage

γδ T cells are present in high numbers among intestinal intraepithelial lymphocytes (IELs) in mice and humans. In addition to immune surveillance for intestinal infection, γδ T cells in IELs are also closely associated with the development and progression of colorectal cancer because intestinal epithelial disturbance and microbial invasion are essential for malignant cell survival and tumor growth.[63,64] γδ T cells are also the main lymphocyte population that infiltrate colorectal cancer,[65,66] but their roles in the development and progression of colorectal cancer remain unclear.

In July 2022, Reis *et al*[67] from Rockefeller University in the United States published a paper in *Science* which revealed that the γδ T-cell subpopulation included both antitumor and protumor subtypes via a parallel analysis of γδ T cells in the infiltrated areas of colorectal cancer and the surrounding tissues of tumors in humans and mice. Using single-cell transcriptomics to analyze the γδ T subtypes and their functions in colorectal tumor tissues and non-tumor tissues, this study found that most γδ T cells in the precancerous colon or non-tumor colon exhibited antitumor activity, while tumor-infiltrating γδ T cells had protumor characteristics. These distinct functional γδ T-cell expression profiles were associated with the usage of different *TCR-Vγδ* genes. γδ T cells played dual roles, and these results lay the foundation for future immunotherapeutic strategies that targeted broad-spectrum γδ T cells.

High levels of glucose impair the antitumor activity of γδ T cells

Increasing blood glucose levels stimulate the proliferation and progression of tumor cells. Moreover, glucose metabolism disorders lead to abnormal tumor monitoring by immune cells in the body. Therefore, the risk of cancer in diabetic patients is greatly increased.[68,69]

In a study published in *Cellular and Molecular Immunology* in August 2022, scientists from the University of Hong Kong, China, found that a high level of glucose might increase the risk of cancer in diabetic patients by inhibiting the antitumor activity of the body's γδ T cells.[70] Moreover, glucose-controlled metabolic reprogramming might improve the antitumor activity of γδ T cells. This study showed that the number and antitumor activity of γδ T cells in diabetic patients were significantly deficient. High glucose induced excessive lactate accumulation in Vγ9Vδ2 T cells, which disrupted the transportion of cytolytic molecules to Vγ9Vδ2 T-cell-tumor synapses by inhibiting adenosine 5′-monophosphate-activated protein kinase (AMPK) activation, resulting in loss of Vγ9Vδ2 T cell antitumor activity.

The results had important implications for the control and prevention of cancer in diabetic patients, and for the first time, these results revealed the important effects of the dysfunction of γδ T-cell cytotoxicity. In addition, metabolic reprogramming mediated by glucose control or metformin therapy might serve as a novel therapeutic target to enhance the antitumor efficacy of γδ T cells.

Latent membrane protein 1 (LMP1) enhances the cytotoxicity of γδ T cells to nasopharyngeal carcinoma by inducing butyrophilin

Nasopharyngeal carcinoma is a diverse cancer that is associated with the carcinogenic EBV. Nasopharyngeal carcinoma has always been treated with chemotherapy and surgery. However, these methods can cause harmful side effects and high recurrence rates.[71–73]

In January 2023, Liu *et al*[74] from Hong Kong Baptist University published research results in *Theranostics* revealing the activation of γδ T cells mediated by the EBV membrane protein LMP1, which could effectively inhibit the growth of nasopharyngeal cancer cells; these findings provided a new idea for immunotherapy of nasopharyngeal cancer. Vδ2 T cells that were amplified *in vitro* had a significant killing effect on nasopharyngeal carcinoma cells. The EBV-targeting probe (L2) P4 reactivated latent EBV in EBV-positive nasopharyngeal carcinoma cells and increased the expression of BTN2A1 and BTN3A1, thereby enhancing the cytotoxicity mediated by adoptive Vδ2$^+$ T cells.

This study demonstrated that a combination of EBV-targeting probes and adoptive Vδ2$^+$ T cells could effectively inhibit the growth of nasopharyngeal carcinoma cells. The elucidation of the mechanism might also lead to new insights and therapeutic targets for nasopharyngeal carcinoma and other "EBV$^+$ tumors."

These are the 2022 basic studies of γδ T cells in tumor immunity, which highlight the pleiotropic functions of γδ T cells in tumor immunity and the dynamic interaction between antitumor and protumor γδ T-cell subsets. Figure 1 summarizes the antitumor and protumor mechanisms of γδ T cells.

Clinical Treatment with γδ T Cells

The first clinical trial related to γδ T-cell immunotherapy started in 2003, and the numbers are increasing yearly. By the end of early 2023, a total of 43 clinical trials were registered in clinical trials [Table 1]. The most recent clinical trial data that were presented at the 49th European Society for Blood and Bone Marrow Transplantation Annual Conference showed that allogeneic γδ T-cell therapy (IN8bio-100) for high-risk or recurrent acute myeloid leukemia after hematopoietic stem cell transplantation showed positive efficacy. γδ T-cell-based immunotherapy has received increasing attention from the medical field worldwide for the first time.

Currently, several companies are developing γδ T-cell-based immunotherapies. These therapies are primarily adoptive cell therapies, in which unmodified or genetically engineered γδ T allogeneic cells are amplified *in vitro* and injected back into the body to treat tumor patients. In addition, other strategies are based on the activation and amplification of γδ T cells *in vivo* or target naturally occurring γδ T cells *in vivo* to recruit them to tumor cells and promote the killing of tumor cells.[6,11,75–79] Next, we will summarize γδ cell-based immunotherapeutic strategies.

Genetically modified chemotherapy-resistant γδ T-cell allotherapy

Chemotherapy is the main means of solid tumor treatment, but it also destroys immune cells. Moreover, lymphocyte clearance is often required before chemotherapy, which can severely inhibit the antitumor immune response, including that of γδ T cells.[80,81] Genetically modified allogeneic natural γδ T cells are protected from chemotherapy-induced damage, allowing them to be simultaneously delivered with chemotherapy to identify and kill remaining tumor cells when the tumor experiences maximal chemotherapy-induced stress.

IN8bio's (Birmingham, Alabama, United States) chemotherapy resistance gene-modified γδ T-cell products include INB-200 and INB-400. INB-200, which is a genetically modified autologous γδ T cell, has been shown to be safe and well tolerated in clinical trials for the treatment of glioblastoma and will soon enter phase 2 clinical trials (Table 2, NCT04165941, https://clinicaltrials.gov/). INB-400, which is a genetically modified allogeneic γδ T-cell product, will be extended to other solid tumor types (Table 2, NCT05664243, https://clinicaltrials.gov/).

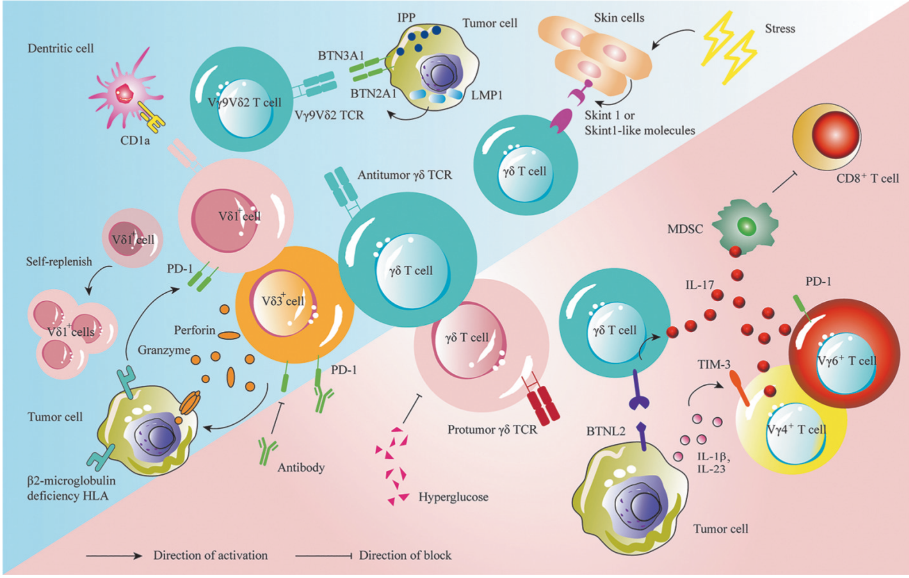

Figure 1: New mechanisms of γδ T cells in anti-tumor or pro-tumor function. Antitumor and protumor γδ T cells are associated with different TCR gene usage. γδ T cells perform tumor surveillance by recognizing Skint1 and Skint1-like molecules. CD1a is a ligand of Vδ1 cells. β2-microglobulin deficiency causes infiltrated Vδ1 and Vδ3 cells to express high levels of PD-1. When PD-1 is blocked, γδ T cells can kill tumor cells without HLA I restriction, such as the release of perforin and granzyme. Vγ9Vδ2 T cells target the accumulation of IPP in cells and produce cytotoxic activity. LMP1 enhances the cytotoxicity of γδ T cells by increasing the expression of BTN2A1 and BTN3A1. Vδ1$^+$ T cells have stem-like characteristics that allow them to self-replenish. BTNL2 of tumor cells reduces cytotoxic CD8+ T cell infiltration by acting on γδ T cells to promote IL-17 secretion and recruit MDSC. Tumor cells secreted IL-1β and IL-23, resulting in high expression of PD-1 and TIM-3 in Vγ6$^+$ cells and Vγ4$^+$ cells, respectively, and secretion of IL-17. Hyperglucose impair the antitumor activity of γδ T cells. BTN3A1: Butyrophilin sub-family 3 member A1; BTN3A2: Butyrophilin sub-family 3 member A2; BTNL2: Butyrophilin-like protein 2; CD: Cluster of differentiation; HLA I: Human leukocyte antigen type I; IL:Interleukin; IPP: Isopentenyl-pyrophosphate; LMP1: Latent Membrane Protein 1; MDSC: Myeloid derived suppressor cells; PD-1: Programmed cell death-1; Skint1: Selection and upkeep of intraepithelial T cells 1; TCR: T cell receptor; TIM-3: T cell immunoglobulin domain and mucin domain-3.

Chimeric antigen receptor (CAR) γδ T-cell therapy

The expression of CAR in T cells enhances the anti-tumor activity of T cells, and it is effective against tumor cells that express the target antigen. CAR T-cell therapies that target CD19 have shown promising results in patients with B-cell malignancies, particularly acute lymphoblastic leukemia, and large B-cell lymphoma.[82–84]

ADI-001 is an allogeneic universal CAR γδ T-cell therapy that targets the B-cell antigen CD20. In December 2022, Adicet Bio (Stanford, California, United States) announced new data from a phase 1 study of ADI-001 in patients with relapsed or refractory B-cell non-Hodgkin lymphoma. The clinical trial results showed that ADI-001 demonstrated good safety and persistence in patients with aggressive non-Hodgkin lymphoma who had been heavily pretreated. Four patients who had previously received autoCD19 CAR T therapy had a 100% complete response rate (Table 2, NCT04735471, NCT04911478).

In addition, this company was also advancing CAR γδ T-cell therapy in solid tumors. ADI-002, which was Adicet Bio's own GPC3-CAR Vδ1 T-cell therapy for GPC3-positive liver cancer, was also in the process of investigational new drug (IND) submission.

Natural, unmodified γδ T-cell allotherapy

γδ T cells exert highly cytotoxic effects on tumor cells without MHC restriction, which sets them apart in the field of antitumor immunity.[6,66] TC BioPharm (Edinburgh, UK) is focused on the discovery, development, and commercialization of natural, unmodified allogeneic γδ T-cell therapies. It is the first clinical study company in oncology to meet International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH) standards.

In March 2022, this company announced the interim results of a phase 1a/2b clinical study of OmnImmune®, its allogenic natural γδ T-cell product, for the treatment of acute myeloid leukemia. Of the four patients who were treated with high doses, two patients achieved a complete response with good safety. Currently, phase 2b/3 clinical trials on OmnImmune® have been initiated. This was the first cryo-thawing/thawing γδ T therapy to enter clinical trials, allowing clinicians to take the product from a storage facility to a specialty pharmacy for thawing and treatment delivery to patients in a short time.

γδ TCR-T-cell therapy

The γδ TCR recognizes and binds to γδ TCR ligands of tumor cells, such as phosphorylated antigens. Therefore, many researchers aim to express the γδ TCR on αβ T cells to endow them with the abilities of natural immune responses and more rapid activation.[85–88] Gadeta (Utrecht, the Netherlands) aims to produce a T cells engineered with defined γδ TCRs (TEG) therapeutic platform for solid tumors.

This company developed a TEG platform for the preparation of γδ-TCR-expressing αβ T cells. These patient-

Table 1: Clinical trial information related to γδ T cells immunotherapy.

Rank	Starting time	Clinical progress	Country	Clinical trial registration No.	Diseases	Immunotherapy based on γδ T cells
1	2003	phase 2	United States	NCT00582790	Kidney cancer	Amplify *in vivo*
2	2008	phase 1	France	NCT00562666	Hepatocellular carcinoma	Unmodified autotherapy
3	2011	phase 1	United States	NCT01404702	Neuroblastoma	Amplify *in vivo*
4	2015	phase 1/2	China	NCT02418481	Breast cancer	Unmodified autotherapy
5	2015	phase 2	China	NCT02425735	Hepatocellular carcinoma	Unmodified autotherapy
6	2015	phase 2	United Kingdom	NCT02459067	Mutiple advanced cancers	Unmodified autotherapy
7	2016	phase 1	United States	NCT02781805	Breast neoplasms	Amplify *in vivo*
8	2017	phase 2	China	NCT02425748	Lung cancer	Unmodified autotherapy
9	2017	phase 2	China	NCT03180437	Pancreatic cancer	Unmodified allotherapy
10	2017	phase 1/2	China	NCT03183206	Breast cancer	Unmodified allotherapy
11	2017	phase 1/2	China	NCT03183219	Liver cancer	Unmodified allotherapy
12	2017	phase 1/2	China	NCT03183232	Lung cancer	Unmodified allotherapy
13	2017	phase 1	Canada	NCT03027102	Acute myeloid leukemia	Allogeneic double negative T cells
14	2018	phase 1	Czechia	NCT03790072	Acute myeloid leukemia	Unmodified allotherapy
15	2019	phase 1	China	NCT04008381	Acute myeloid leukemia	Unmodified allotherapy
16	2019	phase 1/2	China	NCT02585908	Gastric cancer	Unmodified autotherapy
17	2019	phase 1	Malaysia	NCT04107142	Mutiple solid tumors	Anti-NKG2DL allogeneic CAR-γδ T cells
18	2020	phase 1	United States	NCT03533816	Mutiple hematological tumor	Unmodified autotherapy
19	2020	phase 1/2	France	NCT04243499	Mutiple advanced cancers	Target BTN3A activated γδ T cells
20	2020	phase 1	United States	NCT04165941	Glioblastoma	Gene-modified γδ T cells or drug resistant immunotherapy
21	2020	early phase 1	China	NCT04702841	T cell-derived malignant tumors	Anti-CD7 allogeneic CAR-γδ T cells
22	2020	early phase 1	China	NCT04518774	Hepatocellular carcinoma	Unmodified allotherapy
23	2020	early phase 1	China	NCT04696705	Non-Hodgkin's lymphoma, peripheral T cell lymphoma	Unmodified allotherapy
24	2021	phase 1	United States	NCT04688853	Relapsed/refractory multiple myeloma	γδ TCR-T cell therapy
25	2021	phase 1	China	NCT05471323	Acute myeloid leukemia	Unmodified allotherapy
26	2021	phase 1/2	China	NCT04765462	Malignant solid tumors	Unmodified allotherapy
27	2021	phase 1	United States	NCT04735471	Multiple lymphoma	Anti-CD20 allogeneic CAR-γδ T cells
28	2021	phase 1/2	United States	NCT04887259	Multiple hematological tumors	Bispecial antibody
29	2021	phase 1	United States	NCT05001451	Acute myeloid leukemia	Unmodified allotherapy
30	2021	phase 1	United States	NCT05015426	Acute myeloid leukemia	Unmodified allotherapy
31	2021	phase 1/2	China	NCT04764513	Mutiple hematological tumors	Unmodified allotherapy
32	2021	phase 1/2	China	NCT05628545	Advanced hepatocellular carcinoma	Unmodified allotherapy
33	2021	phase 2	Germany	NCT05080790	Leiomyosarcoma	Amplify *in vivo*
34	2022	/	United States	NCT04911478	Multiple lymphoma	Anti-CD20 allogeneic CAR-γδ T cells
35	2022	phase 1	Singapore	NCT05302037	Mutiple solid and hematological tumors	Anti-NKG2DL allogeneic CAR-γδ T cells
36	2022	phase 1/2	France	NCT05307874	Advanced solid tumor	Amplify *in vivo*
37	2022	phase 1	China	NCT05453669	B-cell non-Hodgkin's lymphoma	CD19-CAR-double negative T cells
38	2022	phase 1/2	United States	NCT05369000	Prostate cancer	Bispecial antibody
39	2022	phase 2	United Kingdom	NCT05358808	Acute myeloid leukemia	Unmodified allotherapy
40	2023	phase 1/2	United States	NCT05664243	Glioblastoma	Autologous genetically modified γδ T cells
41	2023	phase 1	United States	NCT05653271	Multiple lymphoma	Allogeneic CD20-conjugated γδ T cell
42	2023	phase 1	United States	NCT05400603	Neuroblastoma	Allogeneic γδ T cells combination drug
43	2023	phase 1	United States	NCT05377827	Mutiple T-cell lymphoma	Anti-CD7 allogeneic CAR-γδ T cells

Clinical trial registration Nos. are from http://clinicaltrials.gov; BNT3A: Butyrophilin sub-family 3 member A; CAR: Chimeric antigen receptor; CD: Cluster of differentiation; NKG2DL: Natural killer group 2D ligand; TCR: T cell receptor.

Table 2: Major γδ T cell therapy companies and their products.

Company	Clinical progress	Clinical trial registration No.	Products	Target	Disease
IN8Bio	phase 1	NCT03533816	INB-100	/	Acute and chronic leukemia, myelodysplastic syndrome
	phase 1	NCT04165941	INB-200	Resistant to alkylation chemotherapy	Adult brain tumor
	phase 2	NCT05664243	INB-400		Glioblastoma
Adicet Bio	phase 1	NCT04735471 NCT04911478	ADI-001	CD20	Non-Hodgkin's lymphoma
	apply for IND	/	ADI-002	GPC3	Hepatocellular carcinoma
TC biopharm	phase 2	NCT03790072 NCT05358808	OmnImmune®	/	Acute myeloid leukemia
	phase 2	NCT04834128	TCB008	/	COVID-19
Gadeta	phase 1	NCT04688853	TEG002	Vγ9Vδ2 TCR	Multiple myeloma
	apply for IND	/			Ovarian cancer
GammaDelta Therapeutics	phase 1	NCT05001451	GDX012	/	Acute myeloid leukemia
CytoMed Therapeutics	phase 1	NCT04107142NCT05302037	CTM-N2D	NKG2DL	Colorectal cancer, gastric cancer, nasopharyngeal carcinoma, prostate cancer, sarcoma, triple negative breast cancer
Lava Therapeutics	phase 1	NCT04887259	LAVA-051	CD1d	Multiple myeloma, acute myeloid leukemia, chronic lymphocytic leukemia
	phase 1	NCT05369000	LAVA-1207	PSMA	Metastatic castration-resistant prostate cancer
ImCheck therapeutics	phase 1	NCT04243499 NCT05307874	ICT01	BTN3A	Solid tumor, adult hematopoietic/lymphoid cancer
Acepodia	phase 1	NCT05653271	ACE1831	CD20	Non-Hodgkin's lymphoma

Clinical trial registration Nos. are from http://clinicaltrials.gov; BNT3A: Butyrophilin sub-family 3 member A; CD: Cluster of differentiation; COVID: Coronavirus disease; GPC: Glypican-3; IND: Investigational new drug; NKG2DL: Natural killer group 2D ligand; PSMA: Prostate-specific membrane antigen; TCR: T cell receptor.

derived αβ T cells retained their function and proliferation capacity, making them ideal vectors for γδ-TCR expression while possessing the broad tumor-targeting capabilities of γδ T cells. Gadeta developed TEG002, which is an autologous αβ T-cell therapy that was modified *in vitro* to express the specific Vγ9 TCR and recognize conformational changes in the CD227 molecules (Table 2, NCT04688853). These engineered cells combined both the powerful effects of αβ T cells and high-affinity TCRs with a wide range of tumor reactivity.

Allogeneic Vδ1⁺ T cell therapy

At present, there are two main subtypes of γδ T cells. The Vγ9Vδ2 subtype is mainly distributed in peripheral blood,[89] while the Vδ1 subtype is a rare type that after maturity, is mainly distributed in epithelial mucosal tissues and has a high incidence of solid tumors, such as skin, digestive tract, respiratory tract, and reproductive tract; this latter cell type can better penetrate tumors and help coordinate the local antitumor immune response.[15,90–92]

GDX012, which is manufactured by GammaDelta Therapeutics(London, United Kingdom), is a treatment based on allogeneic natural Vδ1⁺ T cells. In May 2022, the U. S. Food and Drug Administration approved an IND application for GDX012 to begin a phase I clinical trial to evaluate the safety and efficacy of this treatment in patients with measurable residual disease-positive acute myeloid leukemia (Table 2, NCT05001451). GDX012 also received FDA orphan drug designation for the treatment of acute myeloid leukemia.

γδ T-cell bispecific antibody adapter

In addition to γδ T-cell adoptive immunotherapy, several γδ T-cell-based immunotherapeutic strategies are based on activating a patient's own γδ T cells *in vivo* or targeting γδ T cells to tumor cells and promote their killing function.[93–96]

LAVA Therapeutics (Utrecht, the Netherlands) is a clinical-stage immuno-oncology company in the Netherlands that is focused on the development of γδ T-cell bispecific antibodies. Its Gammabody™ modular platform is a bispecific γδ T-cell adaptor platform that targets and activates Vγ9Vδ2 T cells *in vivo*. In addition to preserving non-specific stress signal recognition, increased tumor antigen-specific recognition can effectively kill tumor cells while preventing non-targeted killing toxicity.

This company has two products in early-stage clinical trials. LAVA-051, which is the first variable domain of heavy chain of heavy-chainantibody(VHH)antibody that targets CD1d, recruits both Vγ9Vδ2 T cells and Invariant natural killer T cells, (iNKT cells) to CD1d-positive tumors and is developed for the treatment of hematologic tumors. In March 2022, This company announced preliminary data from a phase 1/2a clinical trial of LAVA-051, followed in December by updated clinical data showing a favorable safety and tolerability profile (Table 2, NCT04887259). LAVA-1207 is a VHH-Fc antibody that targets the prostate-specific membrane antigen (PSMA). Fc binding prolongs the half-life of the drug and silences the Fc-mediated effect. In 2022, LAVA-1207 was currently being recruited for an open-label, multicenter, phase 1/2a clinical trial for the treatment of metastatic castration-resistant prostate cancer to evaluate its safety, tolerability, pharmacokinetics, pharmacodynamics, immunogenicity, and initial antitumor activity (Table 2, NCT05369000).

Targeting BTN3A-activated γδ T cells in vivo

The precise TCR-targeting mechanism of Vγ9Vδ2 T cells is controversial, but the relevant research has shown that the TCR Vγ9 chain directly recognizes BTN2A1, which is associated with BTN3A1 (CD277), which is another ligand recognized by the TCR Vδ2 chain. Phosphoantigens bind to the intracellular B30.2 domain of BTN3A1, which enables the BTN2A1–BTN3A1 complex to bind to the Vγ9Vδ2 TCR.[97–99]

In September 2022, ImCheck Therapeutics (Paris, France) announced data from a phase 1 dose escalation study of a phase 1/2a clinical trial that studied the use of ICT01 in combination with pembrolizumab to treat solid tumors. ICT01 is a monoclonal antibody against BTN3A, and it binds to BTN3A to specifically activate Vγ9Vδ2 T cells and promote their migration and invasion of tumors. Data showed that this combination therapy enabled a broad antitumor immune response and durable disease control in patients, and this approach showed good safety and had the potential to become a new treatment option for these patients (Table 2, NCT04243499 and NCT05307874).

Antibody conjugated to γδ T cells

Acepodia is an American clinical stage biotechnology company with a unique antibody-coupled cell (ACC) technology platform. Instead of changing the genome of a cell, this technique uses chemical coupling to enable natural immune cells with killer functions to target a specific antigen; this approach combines the precise targeting of monoclonal antibodies with cancer-killing immune cells to generate a potent cell therapy that enhances the binding and killing of low-antigen tumors.[100]

ACE1702 is an anti-HER2 antibody-NK cell-coupled drug candidate that is currently in phase I clinical development for the treatment of HER2-expressing solid tumors. In June 2022, an IND application for ACE1831, which is an allogeneic γδ T-cell candidate with CD20 antibodies that is in development, was approved by the U.S. Food and Drug Administration (FDA) for the initiation of phase 1 multicenter clinical study in patients with non-Hodgkin's lymphoma (Table 2, NCT05653271).

Figure 2 shows the common means by which γδ T cells are used in clinical therapy. As a "new trend" in tumor immunotherapy, γδ T cells have attracted increasing attention and promoted the rapid development of related industries. As a clinical trial has progressed, γδ T-cell therapy has shown good efficacy and safety.[10,13,15]

Among the companies in China, Ruichuang Biology is mainly focused on double-negative T-cell (DNT) therapy, which is a representative γδ T cell-based tumor immunotherapy. DNT cells are a special type of T-cell that do not express CD4 and CD8 molecules. Most DNT cells are γδ T cells.[101] In January 2023, Ruichuang's injectable RC1012 was officially approved by the Center for Drug Review and Evaluation (CDE) of the China National Medical Products Administration to prevent the recurrence of acute myeloid leukemia after allogeneic hematopoietic stem cell transplantation. RC1012 is an unmodified generic DNT immune cell therapy product from a healthy donor without HLA typing.

Challenges and Prospects for the Future

γδ T cells are present in high numbers in epithelial mucosal tissues and exert strong tumor-targeting cytotoxic effects without MHC restriction, and immuno-

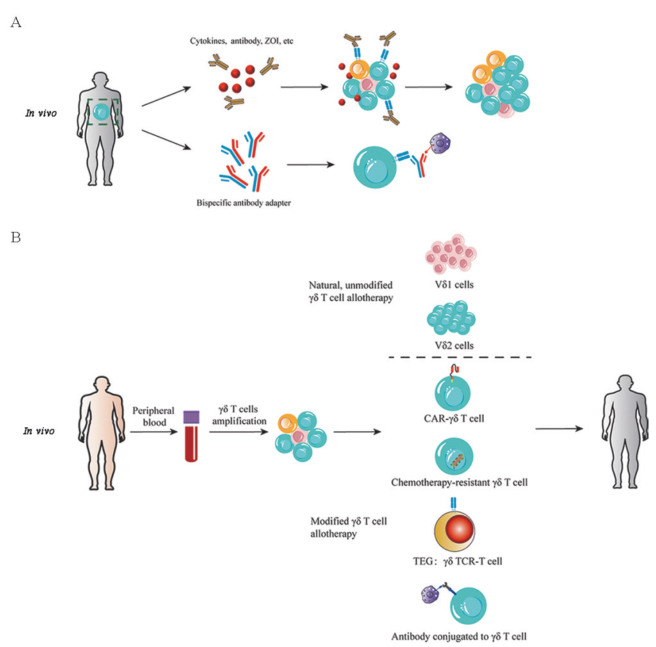

Figure 2: Common immunotherapy methods based on γδ T cells. (A) Immunotherapy *in vivo*: Selective amplification of γδ T cells *in vivo* by cytokines, antibodies, or bisphosphonates, and bispecific antibody connects the γδ T cells to tumor cells. (B) Immunotherapy *in vitro*: The peripheral blood of healthy people is extracted and the γδ T cells are isolated. Natural and unmodified γδ T cell allotherapy is selectively amplifies Vδ1 and Vδ2 cells. Modified allotherapy methods include CAR-γδ T cell, gene editing chemotherapy resistant γδ T cell, γδ TCR-T cell, and antibody conjugated to γδ T cell. CAR: Chimeric antigen receptor; TEG: T cells engineered with defined γδ TCRs;TCR:T cell receptor; ZOI: Zone of inhibition.

therapies based on these cells have received worldwide attention; additionally, the related industry has rapidly developed to accelerate their clinical application. However, we should note that many key questions remain unanswered about γδ T cells due to the lagging of basic research; this has already greatly limited the development of new strategies to enhance the efficacy of γδ T-cell tumor immunotherapeutic strategies and their further clinical application. First, most basic research studies on γδ T cells have been conducted in mice, and the mechanisms involved cannot be directly translated to human γδ T cells due to the large differences between human and mouse γδ T cells. For example, IL-17-secreting γδ T cells have been shown to promote tumor growth in mice. In humans, however, there is currently no evidence of a specific subpopulation of γδ T cells that is dedicated to IL-17 production.[102] Second, an in-depth understanding of the characteristics of human γδ T cells, including their development, differentiation, homing, activation, proliferation, effect, and regulation, is lacking. Most studies have directly applied the known mechanisms of αβ T cells or NK cells to γδ T cells. PD-1, which is an important immune checkpoint molecule that is expressed by αβ T cells, has also been used as a negative regulator of γδ T cells in many studies to enhance the antitumor efficacy of γδ T cells.[103] This approach can greatly accelerate the application of γδ T cells. On the other hand, this approach may restrict future basic research on γδ T cells because perhaps no one will pay attention to the potential presence of specific immune checkpoints in γδ T cells. Third, γδ T cells, which are part of a complex immune regulatory network, do not function alone but interact with other cells *in vivo*. For example, Vγ9Vδ2 T cells assist dentritic cell maturation by producing tumor necrosis factor-α (TNF-α) and Interferon γ (IFN-γ), generating DCs with upregulated costimulatory molecule expression and enhanced cytokine secretion, effectively inducing αβ T-cell proliferation and IFN-γ production.[104] Instead, DCs produce stimulating cytokines, such as IL-12, IL-1β, TNF-α, and IFN-γ, to activate γδ T-cell cytokine production and cytotoxicity.[105,106] Most of the current basic research is fragmented and isolated. An in-depth study elucidating the mechanisms by which γδ T cells interact with other immune cells is critical and will be beneficial for improving the efficacy of γδ T cell-based immunotherapy.

In the future, a fundamental understanding of all the characteristics of γδ T cells compared with αβ T cells and NK cells in terms of their development, differentiation, proliferation, activation, effector function, and exhaustion is needed. It is necessary to further elucidate the γδ TCR repertoire and the mechanism underlying γδ TCR antigen recognition to identify more tumor-associated antigens that are recognized by γδ T cells. The variable functions of different γδ T-cell subsets, such as $IL\text{-}17^+$ γδ T cells and $V\delta1^+$ γδ T cells, and the plasticity in unique regional immune organs and tumor microenvironments need to be further explored. The mechanisms by which γδ T cells interact with other immune cells need to be further elucidated because these dynamic interactions affect the function of γδ T cells in a delicate immune regulatory network. With comprehensive basic research achievements, we can more specifically develop strategies that favor the antitumor activity of γδ T cells *in vivo* and greatly improve the efficacy of γδ T cell-based tumor immunotherapy.

Overall, γδ T cell research is gaining momentum, with clinical trial numbers expected to dramatically increase in the next few years, and there is substantial potential for commercialization. We expect to see more extensive and in-depth basic studies on the characteristics and functions of γδ T cells to enable a more comprehensive understanding of the biological characteristics of this unique, mysterious, and important group of lymphocytes, to guide the improvement of clinical γδ T-cell technology and the expansion of the application range, and to provide novel application strategies and techniques for γδ T cell-based immunotherapy.

Funding

This work was supported by grants from the National Natural Science Foundation of China (Nos. 32270915, 31970843, 82071791, U20A20374, and 81972886), the National Key Research and Development Program of China (No. 2022YFC3602004), the CAMS Initiative for Innovative Medicine (Nos. 2021-I2M-1-005, 2021-I2M-1-035, and 2021-I2M-1-053), Haihe Laboratory of Cell Ecosystem Innovation Fund (No. 22HHXBSS00028), the CAMS Central Public Welfare Scientific Research Institute Basal Research Expenses (No. 3332020035), Changzhou Science and Technology Support Plan (No. CE20215008), and Beijing Municipal Commission of Science and Technology Fund for Innovative Drug (No. Z221100007922040).

Conflicts of interest

None.

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How to cite this article: Zhao YQ, Dong P, He W, Zhang JM, Chen H. γδ T cells: Major advances in basic and clinical research in tumor immunotherapy. Chin Med J 2024;137:21–33. doi: 10.1097/CM9.0000000000002781